

Prasanna Maddila

PHD CANDIDATE | MARL & GAME THEORY

📍 10 Allée des Sciences Appliquées, 31400 Toulouse

☎ (+33) 07 80 83 59 82

🏠 <https://prasanna.maddila.cc>

🐙 [PrasannaMaddila](#)

🌐 [Prasanna Maddila](#)

🏠 [Prasanna Maddila](#)

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[0000-0003-0049-1746](#)

Education

[University of Toulouse](#) | [Lab MIAT, INRAE](#)

Toulouse, France

PHD CANDIDATE

Oct 2023 - Present

- Working on the Nash Equilibrium computation in games of Teams vs. Independent Adversaries, with application to Anti-Poaching.
- Implemented GPU-compatible solvers in JAX for proposed algorithms, and RL environment for application case.
- Project funded by the French-German ANR Project CHIP-GT.

CentraleSupélec

Paris, France

MASTERS IN ENGINEERING (COMPUTER SCIENCE)

August 2021 - 2023

- Specialisation in Theoretical Computer Science, and part of the Research Track (*Filière de Recherche*).
- Student in the Accelerated Engineering Degree Programme.

Mahindra University

Hyderabad, India

BACHELORS' IN TECHNOLOGY, MECHANICAL ENGINEERING

August 2017 - 2021

- Recipient of the University's Student Scholarship for Academic Excellence for 3 years.
- Member of the Organising Team TEDx (2018-19), and Vice-President of the Student Council (2020).

Experience

End-of-Studies Internship

Toulouse, France

MIAT, INRAE

Apr - Sept 2023

- Modelled Anti-Poaching as a Partially Observable Stochastic Game.
- Implemented a simulator using PettingZoo, with various MARL algorithms using RLlib and PyTorch ([1],[2]).

M2 Project

Paris, France

MICS, CENTRALESUPÉLEC

Nov 2022 - Apr 2023

- Worked on the Roommate Matching problem under Rank-Envy-Free matching constraints [3].
- Established complexity results for such matchings, and studied their relation to existing notions.

M1 Internship

Paris, France

MICS, CENTRALESUPÉLEC

May - July 2022

- Worked on Epidemiological models to model the evolution of COVID Variants.
- Implemented numerical schemes and calibrated models using real-world data.

Visiting Student

Shantou, China

UNIVERSITY OF SHANTOU

May - July 2018

- Developed approaches based on genetic algorithms for modelling of Li-ion batteries in Electric Vehicles.
- Worked on data analysis and Experimental Design to minimise physical testing.

Teaching

Deep Learning

Toulouse, France

UNIVERSITY OF TOULOUSE

November 2024 - December 2025

- Created and taught a course on Deep Learning using PyTorch (Years 2024-25 and 2025-26).
- Course offered to final-year Masters' students in [Mathematical Modelling of Ecological Systems](#).
- Course projects: Auto-Encoders for Image Segmentation and training CNN Classifiers for real-world datasets.

Skills

Programming Languages

Python, C, C++17, MATLAB

Libraries

JAX, PyTorch, Ray/RLlib, BenchMARL, NumPy, Pandas, Matplotlib, Pytest, Sphinx

Other Software/Skills

LaTeX, Make, CMake, Bash, Git, CI/CD, Docker (Basics)

Spoken Languages

English (C1/C2), French (B2), Hindi, Telugu

Software

[Anti-Poaching Environment](#)

MIAT, INRAE

- Reference implementation of the Anti-Poaching Environment, implemented using PettingZoo ([1],[2]).
- Implemented MARL algorithms using RLLib to leverage multiple CPUs or GPU using PyTorch backend.
- Simulator implements a formally prescribed Partially Observable Stochastic Game (POSG) model.

[Multi-Adversarial Team Games](#)

MIAT, INRAE

- JAX-based, GPU-compatible implementation of the solver for Multi-Adversarial Team Games [4].
- Implementation scales to 4v9 games with 8 actions each, and can be vectorised to solve multiple games in parallel batches.
- Solver can be installed as a Python package.

[Deep Learning Course \(French\)](#)

UNIVERSITY OF TOULOUSE

- Slides and project material for Teaching course offered to Masters' students (CC-BY-4.0 License).
- Course introduces Perceptrons, Dense Networks, CNNs and RNNs, taught in PyTorch.

Preprints

- Prasanna Maddila, Sabbadin Régis, Meritxell Vinyals. Approximating Nash Equilibria in Finite-Horizon Multi-Adversarial Team Markov Games. 2026. (hal-05570430) (Under Review.)

Publications

- [1] P. Maddila, C. Eric, P. Chabrier, R. Sabbadin, and M. Vinyals, "APE: An Anti-poaching Multi-Agent Reinforcement Learning Benchmark," in *Seventeenth European Workshop on Reinforcement Learning*, 2024. [Online]. Available: <https://openreview.net/forum?id=JSWRnHC93W>
- [2] P. Maddila, R. Sabbadin, and M. Vinyals, "Learning in Competitive-Cooperative Games for Anti-Poaching," in *Workshop on Learning in Games*, 2024.
- [3] B. Coutance, P. Maddila, and A. Wilczynski, "Rank-Envy-Freeness in roommate matchings," in *26th European Conference on Artificial Intelligence (ECAI 2023)*, 2023.
- [4] P. Maddila, R. Sabbadin, and M. Vinyals, "Efficiently Computing Approximate Nash Equilibria in Multi-Adversarial Team Games," in *The 25th International Conference on Autonomous Agents and Multi-Agent Systems*, 2026. [Online]. Available: <https://hal.science/view/index/docid/5450199>
- [5] A. Agrawal, P. Maddila, C. Zhang, B. Perabathini, and S. Lasaulce, "Cooperative energy efficient resource allocation in fast fading interference networks," in *2019 International Conference on Wireless and Mobile Computing, Networking and Communications (WiMob)*, 2019, pp. 276–281.
- [6] S. K. C. Bulusu, P. Maddila, H. Shaiek, and D. Roviras, "HPA linearization for FBMC-OQAM signals with fast convergence-digital predistortion," in *2019 16th International Symposium on Wireless Communication Systems (ISWCS)*, 2019, pp. 133–138.
- [7] L. Yun *et al.*, "An integrated framework for minimization of inter lithium-ion cell temperature differences and the total volume of the cell of battery pack for electric vehicles," *Energy Storage*, vol. 1, no. 2, p. e41, 2019.
- [8] L. Yun *et al.*, "Maximization of extraction of Cadmium and Zinc during recycling of spent battery mix: An application of combined genetic programming and simulated annealing approach," *Journal of Cleaner Production*, vol. 218, pp. 130–140, 2019.